

Why a Time-Dependent Neural-Network Potential Correction Cannot Recover h_{true} Under an Initial-PDF Mismatch

Setup

We back-integrate observed stellar coordinates (z_i, w_i) at observation time t_{obs} to initial coordinates $(z_i(0), w_i(0))$ using a model gravitational potential

$$\Phi_{\text{model}}(z, t) = \Phi_{\text{static}}(z; h) + V_{\text{nn}}(z, t), \quad (1)$$

where Φ_{static} is the fixed baryon + dark-matter-halo contribution parametrised by the dark-matter scale height h , and $V_{\text{nn}}(z, t)$ is an arbitrary neural-network potential correction. The likelihood assumes a Gaussian initial distribution,

$$f_0(z, w) = \mathcal{N}(0, \sigma_{z, \text{fit}}) \times \mathcal{N}(0, \sigma_{w, \text{fit}}), \quad (2)$$

and the maximum-likelihood estimate minimises

$$\text{NLL}(h, V_{\text{nn}}) = \sum_i \left[\frac{1}{2} \frac{z_i(0)^2}{\sigma_{z, \text{fit}}^2} + \frac{1}{2} \frac{w_i(0)^2}{\sigma_{w, \text{fit}}^2} \right] + \text{const.} \quad (3)$$

over (h, V_{nn}) jointly. The data was generated with $(\sigma_{z, \text{true}}, \sigma_{w, \text{true}})$. The “PDF mismatch” we study is the case in which $\sigma_{w, \text{fit}} < \sigma_{w, \text{true}}$ (the argument is identical for $\sigma_{z, \text{fit}} < \sigma_{z, \text{true}}$).

Claim

Claim. *No $V_{\text{nn}}(z, t)$ — regardless of architecture, time-dependence, network capacity, or physical constraints (positivity, smoothness, finite mass, etc.) — can move the joint MLE to $h^* = h_{\text{true}}$.*

The argument below proceeds in five short steps. The conclusion is *structural*: the bias lives in the loss function itself, not in the model space, and no enrichment of the model space can compensate for that.

Step 1: At the truth, the loss is large, not small

Plug in the correct model: $h = h_{\text{true}}$, $V_{\text{nn}} \equiv 0$. The dynamics are then exactly those used to generate the data, and the back-integration is exact:

$$w_i(0) \sim \mathcal{N}(0, \sigma_{w, \text{true}}), \quad z_i(0) \sim \mathcal{N}(0, \sigma_{z, \text{true}}). \quad (4)$$

The expected per-star w -term in the NLL is

$$\mathbb{E} \left[\frac{1}{2} \frac{w(0)^2}{\sigma_{w, \text{fit}}^2} \right] = \frac{1}{2} \frac{\sigma_{w, \text{true}}^2}{\sigma_{w, \text{fit}}^2}. \quad (5)$$

When $\sigma_{w, \text{fit}} < \sigma_{w, \text{true}}$, this is much larger than the $\frac{1}{2}$ that would obtain if the PDF were correct. The truth is *not* a stationary point of the loss; the gradient $\partial \text{NLL} / \partial h$ at $(h_{\text{true}}, 0)$ is non-zero, and the optimiser will move *away* from the truth in order to reduce the loss.

Step 2: The loss minimum lies wherever the back-integrated w -distribution is narrow

The optimiser wants the back-integrated initial-velocity variance to satisfy

$$\langle w(0)^2 \rangle \approx \sigma_{w,\text{fit}}^2 \quad (6)$$

so that the squared term in (3) is of order one. Two *physically independent* mechanisms in the model can produce such a narrow $w(0)$ distribution:

(a) Drift in h .

Lowering h increases the magnitude of the anharmonic z^4/h^2 correction in $\Phi_{\text{DM}}(z)$. High-amplitude orbits become “stiffer” — their back-integration compresses them more — so the distribution of $w(0)$ is narrower.

(b) A positive-curvature V_{nn} .

Adding a midplane mass concentration produces a quadratic correction $V_{\text{nn}}(z) \approx \frac{1}{2} \Delta\omega^2 z^2$ near the midplane. This raises the effective harmonic frequency at $z = 0$, which in turn modifies the back-integration so that $w(0)$ is narrower. A flexible NN finds this very easily.

The MLE landscape is therefore *not* point-like at any single minimum; it has a continuous ridge of nearly-degenerate solutions parametrised by trade-offs between (a) and (b). The truth $(h_{\text{true}}, 0)$ lies *off* this ridge. Adam picks a point on the ridge based on initialisation and learning rates, but it will not, in general, land at the truth.

Step 3: Time-dependence cannot help — it can only enlarge the degeneracy

A time-dependent correction $V_{\text{nn}}(z, t)$ is a strict superset of $V_{\text{nn}}(z)$:

- Every static V_{nn} that absorbs the PDF bias is still available as a t -constant member of the larger class.
- Additional time-dependent modes — for example, periodic stiffening of the well at orbital frequencies — can produce *additional* mechanisms that narrow the back-integrated $w(0)$ distribution.

The dimension of the degenerate ridge grows. Time-dependence *cannot remove* any direction; it can only add more low-NLL configurations to choose from. A time-dependent NN therefore cannot do better than the static NN — it can only do equally badly or worse.

Step 4: Physical constraints (e.g. $\rho_{\text{nn}} \geq 0$ via Poisson) do not remove the relevant ridge

A natural physical restriction is to require that the NN’s potential correspond to a non-negative mass distribution:

$$\frac{\partial^2 V_{\text{nn}}}{\partial z^2} = 4\pi G \rho_{\text{nn}}(z) \geq 0. \quad (7)$$

This forces V_{nn} to be everywhere convex. It successfully blocks one direction of bias-absorption — concave corrections like $-z^4/h^2$ that would mimic h -decreases by reshaping the anharmonic structure of the potential.

But it does *not* block mechanism (b). A constant positive density near midplane gives a quadratic $V_{\text{nn}}(z) \approx \text{const} \cdot z^2$, which is convex — fully allowed under $\rho_{\text{nn}} \geq 0$. This is exactly

the midplane-mass-concentration that narrows $w(0)$. The bias-absorbing direction therefore *survives* the positivity constraint. The ridge persists.

Any constraint we can write down on the dynamics-side model can only carve along the loss surface. It cannot move the surface itself to a place where $(h_{\text{true}}, 0)$ is the optimum, because that is not what the constraint does.

Step 5: The bias is in the loss function, not the model

The structural obstruction is this. The loss

$$\text{NLL}(h, V_{\text{nn}}) \sim \sum_i \frac{w_i(0; h, V_{\text{nn}})^2}{\sigma_{w,\text{fit}}^2} + \dots \quad (8)$$

has $\sigma_{w,\text{fit}}$ **frozen at the wrong value**. The minimum of any such loss over any model class — potential corrections, time-dependent corrections, physics-informed corrections, neural-network corrections, all of them — sits at the model point that produces $\langle w(0)^2 \rangle \approx \sigma_{w,\text{fit}}^2$. By assumption that is not the truth, because the truth produces $\langle w(0)^2 \rangle \approx \sigma_{w,\text{true}}^2 > \sigma_{w,\text{fit}}^2$.

No flexibility on the model side can change where the loss function's minimum sits, because the loss function is defined by the wrong $\sigma_{w,\text{fit}}$.

A NN potential correction is a tool for redistributing flexibility *within* the dynamics. The bias does not live in the dynamics; it lives in the loss function, via the assumed initial distribution. The tool is pointed at the wrong target.

Corollary: What *can* recover h_{true}

Only two strategies can succeed.

(1) Change the loss function

Make $\sigma_{w,\text{fit}}$ (and $\sigma_{z,\text{fit}}$) **free parameters** of the fit. The joint MLE then minimises over $(h, \sigma_{z,\text{fit}}, \sigma_{w,\text{fit}}, \dots)$, and the joint minimum is at $(h_{\text{true}}, \sigma_{z,\text{true}}, \sigma_{w,\text{true}}, \dots)$. This is the structurally correct fix: the bias was that the loss had a wrong frozen constant; allowing that constant to float removes the bias at its source.

(2) Add information beyond the per-star NLL

Cross-validation against held-out stars, multiple datasets at different t_{obs} , an external prior on h , observed spatial structure of $\sigma_w(z)$, etc. Each of these adds a constraint that the wrong- σ_w bias cannot satisfy simultaneously, because the bias produces different signatures at different datasets or scales.

Where a NN can usefully live

A NN solution to the PDF-mismatch problem is to put NN flexibility where the bias actually lives — in the initial PDF itself. Replace the rigid Gaussian $f_0(z, w)$ with a NN-parametrised density (e.g. a normalising flow over (z, w) , or its parametric special case: a Gaussian whose parameters are learnable, or a Gaussian mixture). The likelihood becomes

$$\text{NLL} = - \sum_i \log f_{0,\text{NN}}(z_i(0), w_i(0)). \quad (9)$$

This NN can absorb the PDF mismatch but *cannot shift orbits*, so it has no leverage on h other than the structural information in the back-integration. The joint MLE then recovers $h \approx h_{\text{true}}$ together with the correct f_0 .

One-sentence summary

A NN potential correction adds knobs to the *dynamics*, but the bias from a wrong initial PDF lives in the *loss function* — so the dynamics-side knobs can never reach the truth, while a NN that parametrises the initial PDF itself can.